

Interfacial Phenomena

Interfacial phenomena

- **Interface** is the boundary between two phases.
- *Surface* is an interface between liquid and gas or solid and gas.

Surface/interfacial tension

- In the bulk portion of each phase, molecules are attracted to each other equally in all directions, such that no resultant forces are acting on any one molecule.
- At the boundary between phases, however, molecules, are acted upon unequally because they are in contact with other molecules exhibiting different forces of attraction.
- Thus, molecules situated at the interface experience interaction forces dissimilar to those experienced in the bulk phase.

Surface/interfacial tension (cont'd)

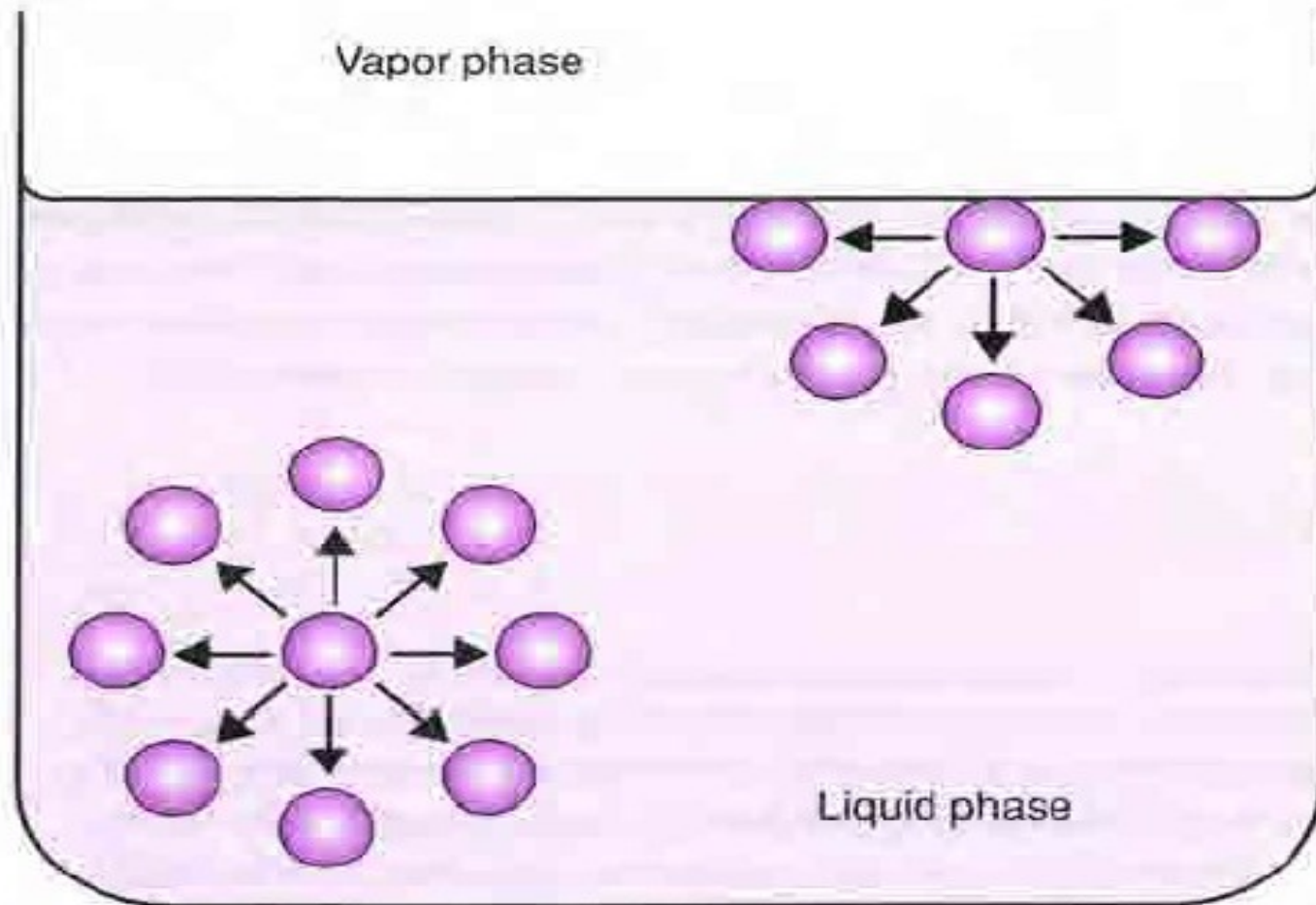


Fig. Representation of the unequal attractive forces acting on molecules at the surface of a liquid as compared with molecular forces in the bulk of the liquid.

Surface/interfacial tension (cont'd)

- In liquid systems such unbalanced forces can be satisfied by spontaneous movement of molecules from the interface to the bulk phase.
- This leaves fewer molecules per unit area at the interface (greater intermolecular distance) and reduces the actual contact area between dissimilar molecules.

Surface/interfacial tension (cont'd)

- Any attempt to reverse this process by increasing the area of contact between phases – that is, bringing more molecules into the interface – causes the interface to resist expansion and behave as though it is under tension everywhere in a tangential direction.

Surface/interfacial tension (cont'd)

- The force of this tension per unit length of interface is called **surface tension /interfacial tension**.
- In the centimeter-gram-second (cgs) system, surface tension is expressed **dyne/cm**.
- In the SI system, interfacial tension is expressed in **milli Newton/meter (mN/m)**.

Example on formation of surface tension

- ✓ A soap film is formed over the area of **ABCD** and can be stretched by applying a force **f** (such as hanging a mass) to the movable bar length **L** , which acts against surface tension of the soap film.

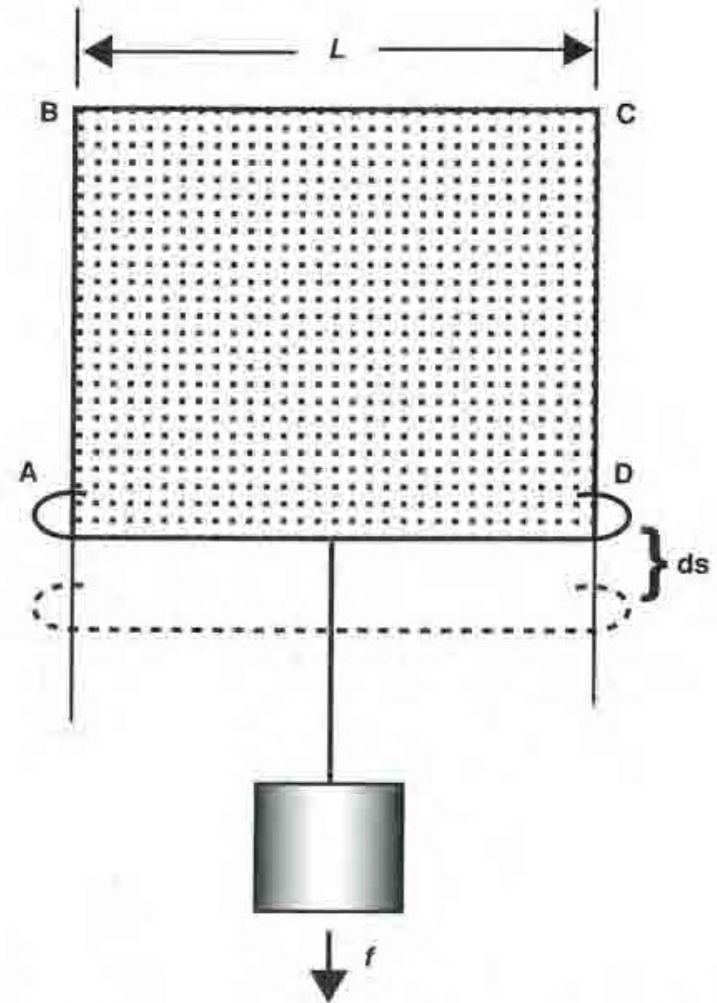


Fig. 15-3. Wire-frame apparatus used to demonstrate the principle of surface tension.

Example on formation of surface tension

- ✓ When the mass is removed, the film will contract owing to its surface tension.
- ✓ The surface tension, γ , of the solution forming the film is the function of the force that must be applied to break the film over the length of the movable bar.

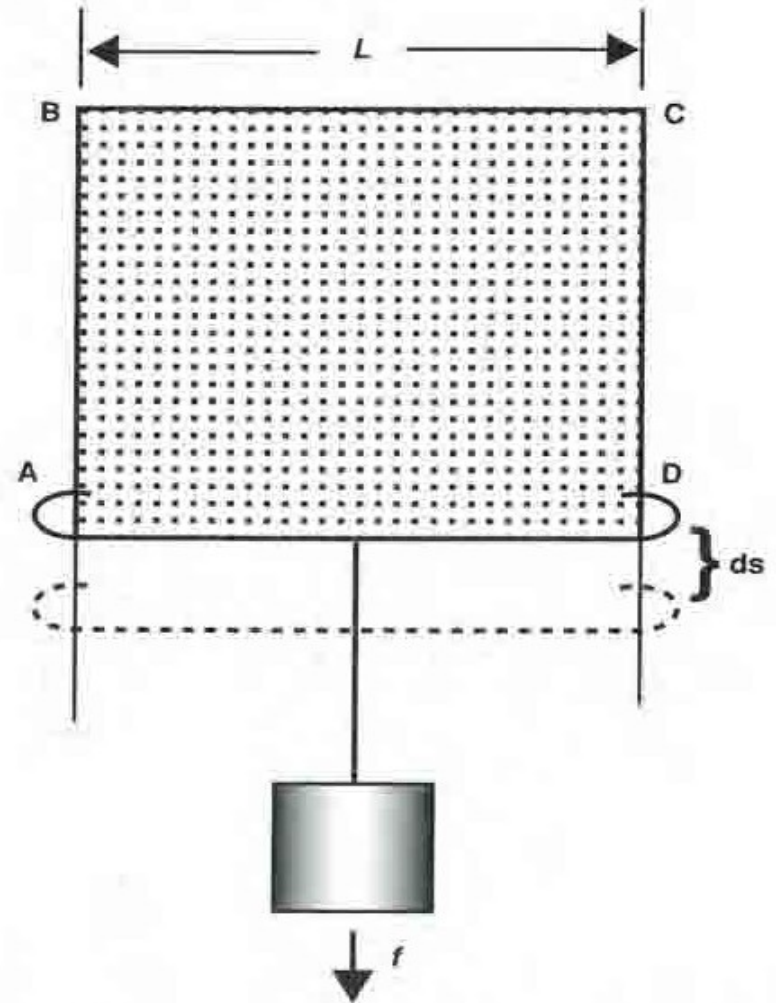


Fig. 15-3. Wire-frame apparatus used to demonstrate the principle of surface tension.

Example on formation of surface tension

- ✓ The soap film has two liquid-gas interfaces. Hence, total length of contact becomes **2L**.

Thus, surface tension

$$\gamma = \frac{f_b}{2L}$$

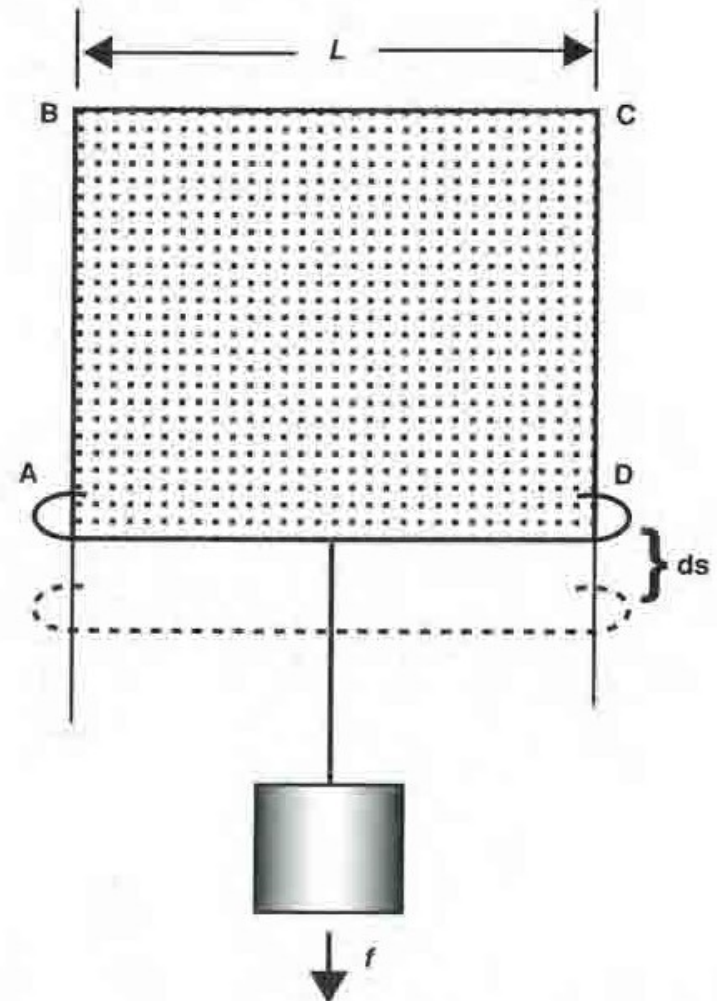


Fig. 15-3. Wire-frame apparatus used to demonstrate the principle of surface tension.

Calculating surface tension of water

- Example,
if the length of the bar, L (in the previous slide), is 5 cm and the mass required to break a soap film is 0.5 g, what is the surface tension of soap solution?

$$\gamma = \frac{0.5 \text{ g} \times 981 \text{ cm/sec}^2}{2 \times 5 \text{ cm}} = 49 \text{ dynes/cm}$$

Surface tension and interfacial tension (against water) at 20 °C.

TABLE 15-2

SURFACE TENSION AND INTERFACIAL TENSION (AGAINST WATER) AT 20°C*

Substance	Surface Tension (dynes/cm)	Substance	Interfacial Tension (dynes/cm)
Water	72.8	Mercury	375
Glycerin	63.4	<i>n</i> -Hexane	51.1
Oleic acid	32.5	Benzene	35.0
Benzene	28.9	Chloroform	32.8
Chloroform	27.1	Oleic acid	15.6
Carbon tetrachloride	26.7	<i>n</i> -Octyl alcohol	8.52
Caster oil	39.0	Caprylic acid	8.22
Olive oil	35.8	Olive oil	22.9
Cottonseed oil	35.4	Ethyl ether	10.7
Liquid petrolatum	33.1		

*From P. Becher, *Emulsions: Theory and Practice*, 2nd Ed., Reinhold, New York, 1962, and other sources.

Measurement of surface/interfacial tension

□ Capillary rise method

- When a capillary tube is placed in a liquid contained in a beaker, the liquid generally rises up the tube a certain distance.
- Because the force of adhesion between the liquid molecules and the capillary wall is greater than the cohesion between the liquid molecules, the liquid is said to *wet* the capillary wall, spreading over it and rising in the tube.

Capillary rise method.....(cont'd)

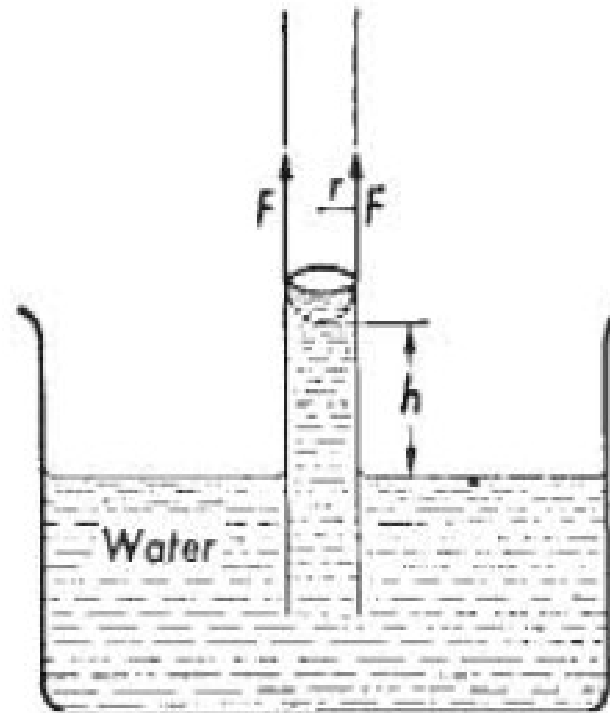


Figure Capillary rise for a liquid exhibiting 0° contact angle. (From Semat H. *Fundamentals of Physics*, 3rd ed. New York: Holt Reinhart Winston, 1957.)

Capillary rise method.....(cont'd)

- By measuring this rise in a capillary, it is possible to determine the surface tension of the liquid.
- It is not possible, however, to obtain interfacial tensions using the capillary rise method.

Capillary rise method.....(cont'd)

- To examine more closely the factors giving rise to the phenomenon of capillarity, consider the case of a liquid that rises to a height, h , above the bulk liquid in a capillary having a radius, r (see next slide).
- If the contact angle of water on glass is 0 , a force, F , will act upward and vertically along the circle of liquid – glass contact.

Capillary rise method.....(cont'd)

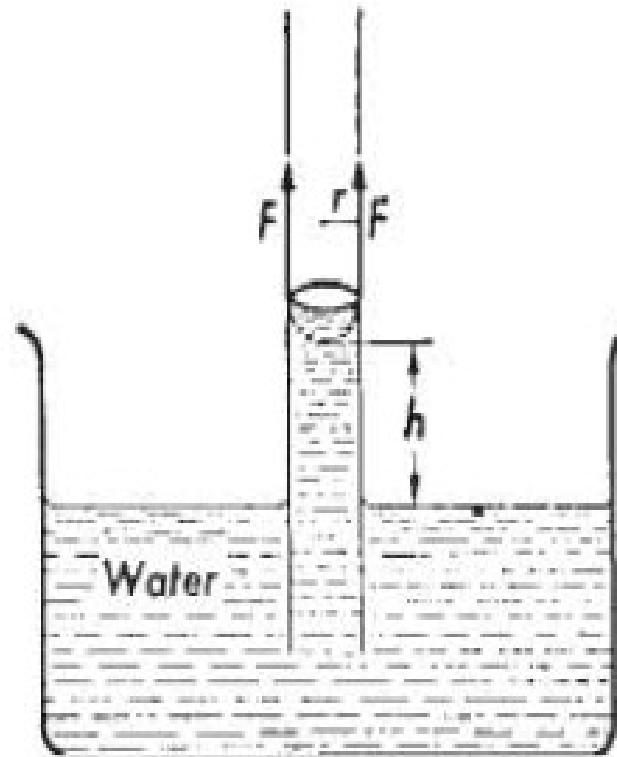


Figure Capillary rise for a liquid exhibiting 0° contact angle. (From Semat H. *Fundamentals of Physics*, 3rd ed. New York: Holt Reinhart Winston, 1957.)

Capillary rise method.....(cont'd)

- Based upon the definition of surface tension, this force will be equal to the surface tension, Υ , multiplied by the circumference of the circle, $2\pi r$,
- Thus, $F = \Upsilon 2\pi r$(1)
- This upward force must support the column of water, and because the mass, m , of the column is equal to the density, ρ , multiplied by the volume of the column, $\pi r^2 h$, the force W opposing the movement upward will be

$$w = mg = \pi r^2 \rho gh$$
.....(2)

Capillary rise method.....(cont'd)

- Equating the two forces at equilibrium, gives,

$$\pi r^2 \rho g h = \gamma 2\pi r$$

so that $\gamma = \frac{r \rho h g}{2}$

Capillary rise method.....(cont'd)

- Example,

a sample of chloroform rose to a height of 3.67 cm at 20 °C in a capillary tube having an inside radius of 0.01 cm. What is the surface tension of chloroform at this temperature?

The density of chloroform is 1.476 g/cm³.

$$\gamma = \frac{1}{2} \times 0.01 \text{ cm} \times 3.67 \text{ cm} \times 1.476 \text{ g/cm}^3 \times 981 \text{ cm/s}^2$$

$$\gamma = 26.6 \text{ g/s}^2 = \underline{26.6 \text{ dyne/cm}}$$

Capillary rise method.....(cont'd)

- If the contact angle of liquid is not **zero(0)** (see next slide), the same relationship may be developed, except the vertical component of **F** which opposes the weight of the column is **Fcosθ**, and, therefore,

$$h = \frac{r \rho g}{2 \cos \theta}$$

Capillary rise method.....(cont'd)

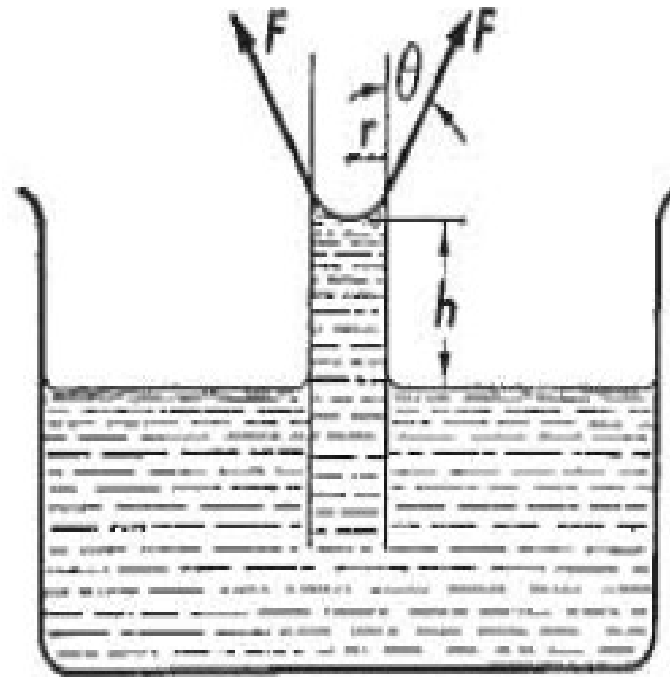


Figure 20-8. Capillary rise for a liquid exhibiting a contact angle, θ , that is greater than 0° but less than 90° . (From Semat H. *Fundamentals of Physics*, 3rd ed. New York: Holt Reinhart Winston, 1957.)

Surface free energy

- To move a molecule from the bulk to the surface, work needs to be done against the force of tension.
- In other words, each molecule at the surface of liquid possesses a certain excess of potential energy as compared to the molecules in the bulk of the liquid.

Surface free energy

- The higher the surface of the liquid, the more molecules have this excessive energy.
- If the surface of the liquid increases (e.g., when water is broken into a fine spray), the energy of the liquid also increases.
- Because this energy is proportional to the size of free surface, it is called **surface free energy**.

Surface free energy (cont'd)

- Each molecule of the liquid, has a tendency to move inside the liquid from the surface; therefore, the liquid takes form with minimal free surface and with minimal surface energy.
- For example, liquid droplets tend to assume a spherical shape because a sphere has the smallest surface area per unit volume.

Surface free energy (cont'd)

- The relationship between interfacial tension and surface free energy can be expressed by the following equation.

$$\Delta G = \gamma \times \Delta A$$

Surface free energy (cont'd)

$$\Delta G = \gamma \times \Delta A$$

Where, ΔG is change in surface free energy, ΔA is change in surface area and γ is surface tension.

- The cgs unit of surface free energy is **ergs (dyne x cm)**.

Surface free energy (cont'd)

Example,

- What is the change in surface energy when 0.5 g of mass hang on 5 cm bar extends the soap film bar by 1 cm?

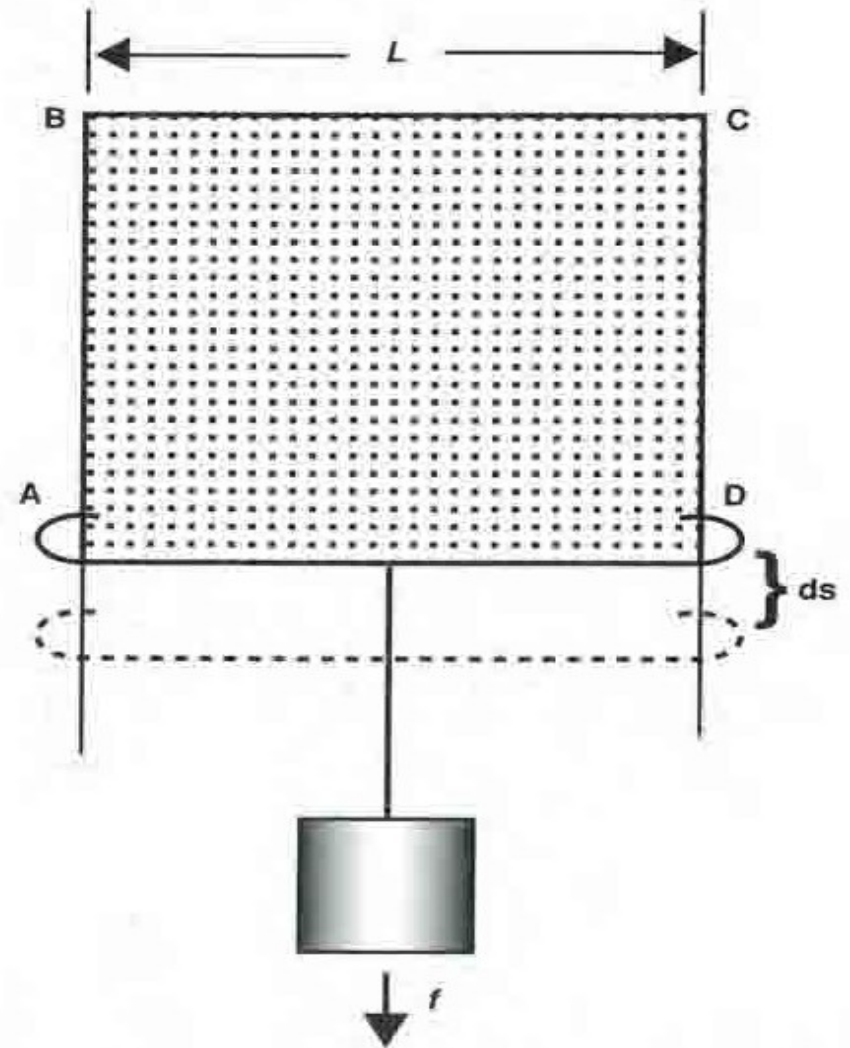


Fig. 15-3. Wire-frame apparatus used to demonstrate the principle of surface tension.

Spreading

- If a small quantity of an immiscible liquid is placed on a clean surface of a second liquid, it may spread to cover the surface with a film or remain as a drop or lens.
- Which of the two applies depends on the achievement of a state of minimum free energy.

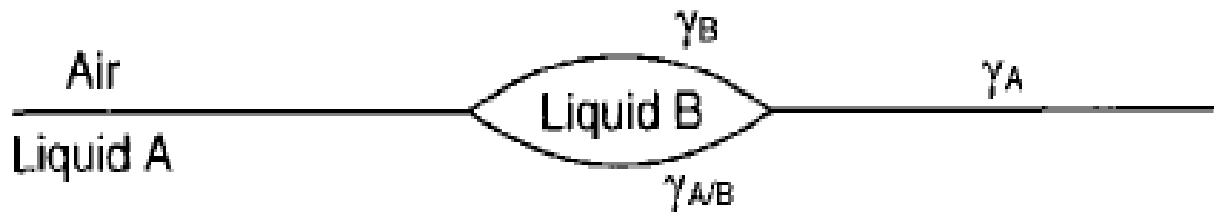


Fig. Spreading of one liquid on another.

Spreading (cont'd)

- Whether an immiscible liquid lays as a lens or spreads over the other liquid, an interface exists.
- Spreading occurs when adhesive forces between spreading molecules and sublayer molecules exceeds cohesive forces between spreading molecules.

Spreading (cont'd)

- The ability of one liquid to spread over another can be assessed in terms of the **spreading coefficient (S)**.
- The spreading coefficient (**S**) of liquid B over liquid A:

$$S = \gamma_A - (\gamma_B + \gamma_{A/B})$$

➤ When S is **positive** spreading occur.

Spreading (cont'd)

- Example,
- ✓ if the surface tension of water is 72.8 dyne/cm at 20 °C, the surface tension of benzene is 28.9 dyne/cm and the interfacial tension between benzene and water is 35 dyne/cm, what is the initial spreading coefficient of benzene on water? Also calculate the spreading coefficient of water on benzene

Spreading (cont'd)

- The spreading coefficient of a liquid may change over time. (i.e., **after equilibration, $+S$ may change to lower $+S$ or $-S$**).
- At this condition, the spreading liquid becomes saturated with the sublayer liquid and vice versa.

Spreading (cont'd)

- Example,
 - ✓ if the surface tension of water is 72.8 dyne/cm at 20 °C, the surface tension of benzene is 28.9 dyne/cm and the interfacial tension between benzene and water is 35 dyne/cm. Following equilibration, the surface tension of water is 62.2 dyne/cm and that of benzene is 28.8 dyne/cm. What is the final spreading coefficient of benzene on water?

Adsorption at liquid interfaces

- Certain molecules and ions, when dispersed in liquid, move of their own accord to the interface.
- Their concentration at the interface then exceeds their concentration in the bulk of the liquid.
- Obviously, the surface free energy and surface tension of the system are automatically reduced.

Adsorption at liquid interfaces

- A phenomenon, where the added molecules are partitioned in favor of the interface, is termed **adsorption**, or, more correctly, **positive adsorption**.

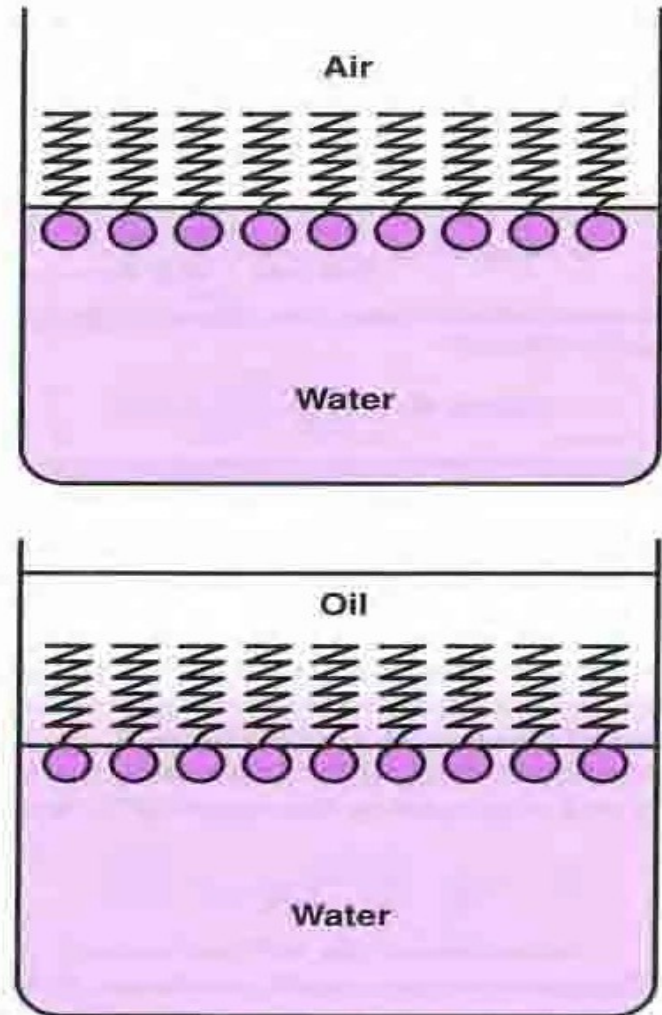


Fig. 15-10. Adsorption of fatty acid molecules at a water–air interface (upper panel) and a water–oil interface (lower panel).

Adsorption at liquid interfaces

- Other materials (e.g., inorganic electrolytes) are partitioned in favor of bulk of aqueous liquids, leading to negative adsorption and a corresponding increase in surface free energy and surface tension.

Adsorption at liquid interfaces (cont'd)

- Adsorption should not be confused with absorption.
- The former is solely a surface effect, whereas in absorption, the liquid or gas being absorbed penetrates into the capillary spaces of the absorbing medium.
 - The taking up of water by sponge is *absorption*.
 - The concentration of alkaloid molecules on the surface of activated charcoal is *adsorption*.

Surface active agents (surfactants)

- Molecules and ions that are adsorbed at interfaces are termed *surface-active agents (surfactants)*.

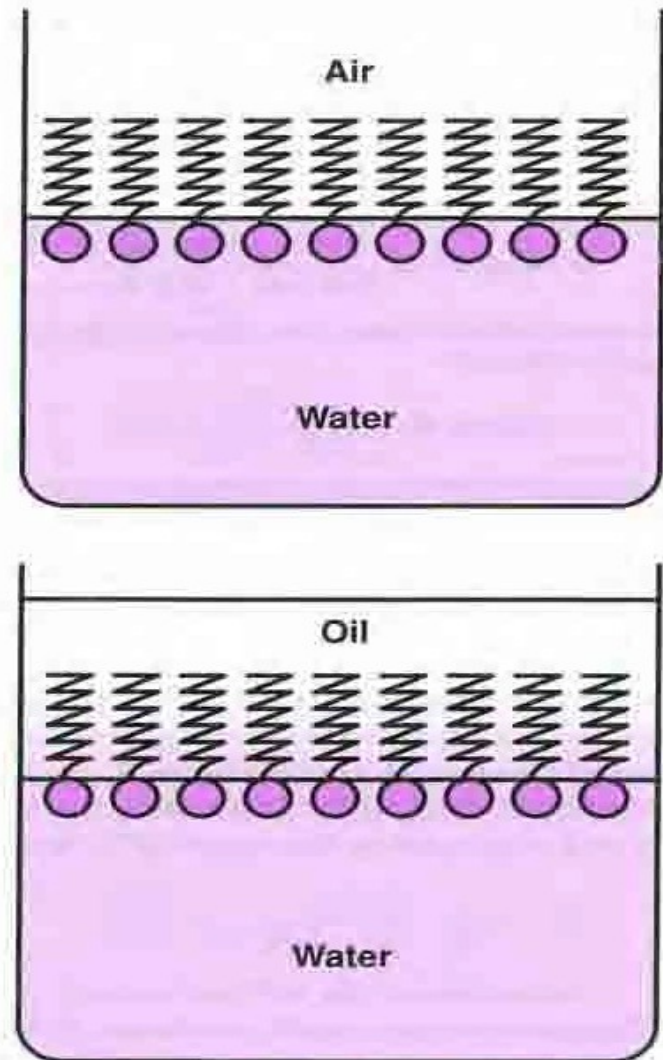


Fig. 15-10. Adsorption of fatty acid molecules at a water-air interface (upper panel) and a water-oil interface (lower panel).

Surface active agents (surfactants)

- An alternative term is *amphiphile*, which suggests that the molecule or ion has a certain affinity for both **polar** and **non polar** solvents.

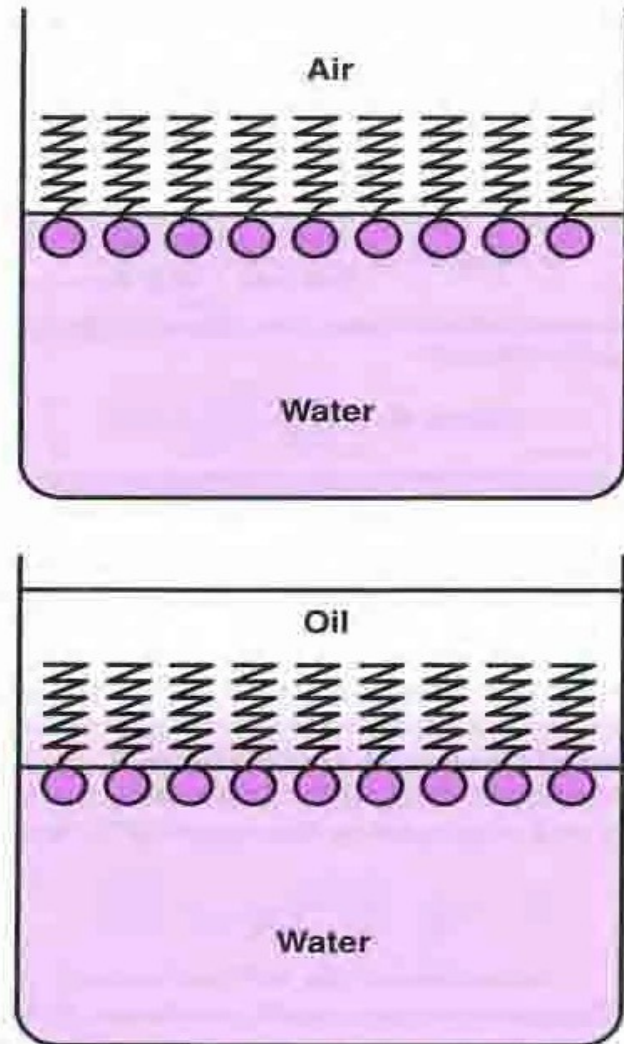


Fig. 15-10. Adsorption of fatty acid molecules at a water-air interface (upper panel) and a water-oil interface (lower panel).

Surface active agents (surfactants)

- Depending on the number and nature of the polar and non polar groups present, the amphiphile may be predominantly **hydrophilic (water-loving)** or **lipophilic (oil-loving)**, or reasonably **well balanced** between these two extremes.
- For example, **straight-chain alcohols, amines,** and **organic acids** are amphiphiles that change from being predominantly hydrophilic to lipophilic as the number of carbon atoms in the alkyl chain increased.

Surface active agents (cont'd)

- If a molecule is too hydrophilic, it remains within the body of aqueous phase and exerts no effect at the interface.
- Likewise, if it is too lipophilic, it dissolves completely in the non-polar phase and little appears at the interface.

The HLB System

- **Griffin** devised an arbitrary scale of values to serve as a measure of the **hydrophilic-lipophilic balance (HLB)** of surface active agents.
- The HLB scale is a numerical scale extending from **1** to approximately **50**.
- The more hydrophilic surfactants have high HLB numbers (in excess of 10).
- Whereas, surfactants with HLB numbers from 1 to 10 are considered to be lipophilic.

The HLB scale...

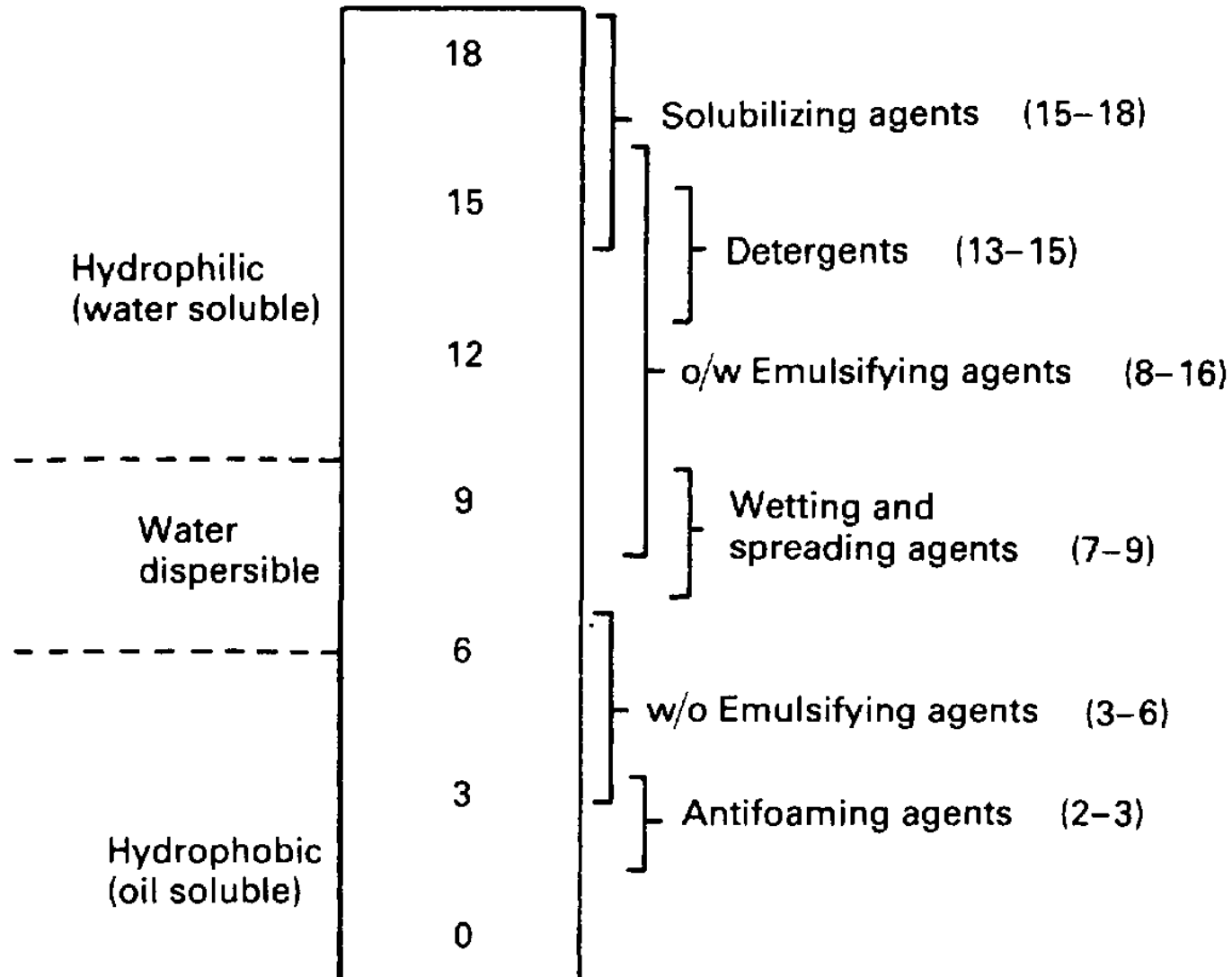


TABLE 15-5**HYDROPHILIC-LIPOPHILIC BALANCE (HLB) VALUES OF SOME AMPHIPHILIC AGENTS**

Substance	HLB
Oleic acid	1
Polyoxyethylene sorbitol beeswax derivative (G-1706)	2.0
Sorbitan tristearate	2.1
Glyceryl monostearate	3.8
Sorbitan monooleate (Span 80)	4.3
Diethylene glycol monostearate	4.7
Glyceryl monostearate, self-emulsifying (Tegin)	5.5
Diethylene glycol monolaurate	6.1
Sorbitan monolaurate (Span 20)	8.6
Polyethylene lauryl ether (Brij 30)	9.5
Gelatin (Pharmagel B)	9.8
Methyl cellulose (Methocel 15 cps)	10.5
Polyoxyethylene lauryl ether (G-3705)	10.8
Polyoxyethylene monostearate (Myrj 45)	11.1
Triethanolamine oleate	12.0
Polyoxyethylene alkyl phenol (Igepal Ca-630)	12.8
Polyethylene glycol 400 monolaurate	13.1
Polyoxyethylene sorbitan monooleate (Tween 80)	15.0
Polyoxyethylene sorbitan monolaurate (Tween 20)	16.7
Polyoxyethylene lauryl ether (Brij 35)	16.9
Sodium oleate	18.0
Potassium oleate	20
Sodium lauryl sulfate	40

HLB calculations

- The HLB of nonionic surfactants (**glycerol esters, polysorbates and sorbitan fatty acid esters**) may be obtained from the equation

$$\text{HLB} = 20\left(1 - \frac{S}{A}\right)$$

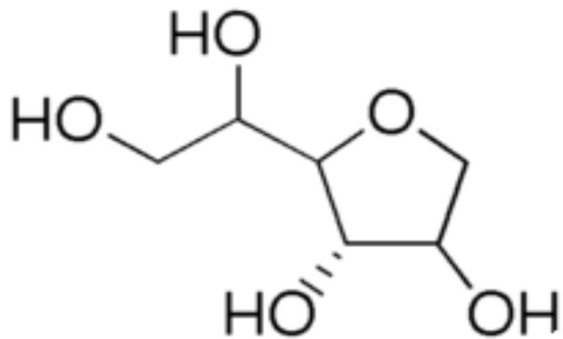
S is saponification value of the ester and

A is acid value of the fatty acid

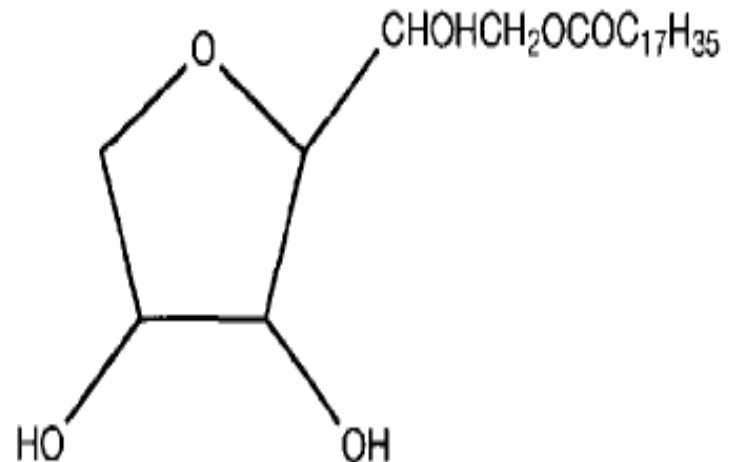
- **S is the weight in mg of KOH required to neutralize the free fatty acids and saponify the esters in 1 g of fatty material.**
- **A is the weight in mg of KOH required to neutralize the free acid in 1 g of fatty material.**

Sorbitan fatty acid esters (Spans)

- These are produced by the esterification of one or more of the hydroxyl groups of sorbitan with either **lauric acid** ($C_{11}H_{23}COOH$), **oleic acid** ($C_{17}H_{33}COOH$), **palmitic acid** ($C_{15}H_{31}COOH$) or **stearic acid** ($C_{17}H_{35}COOH$)



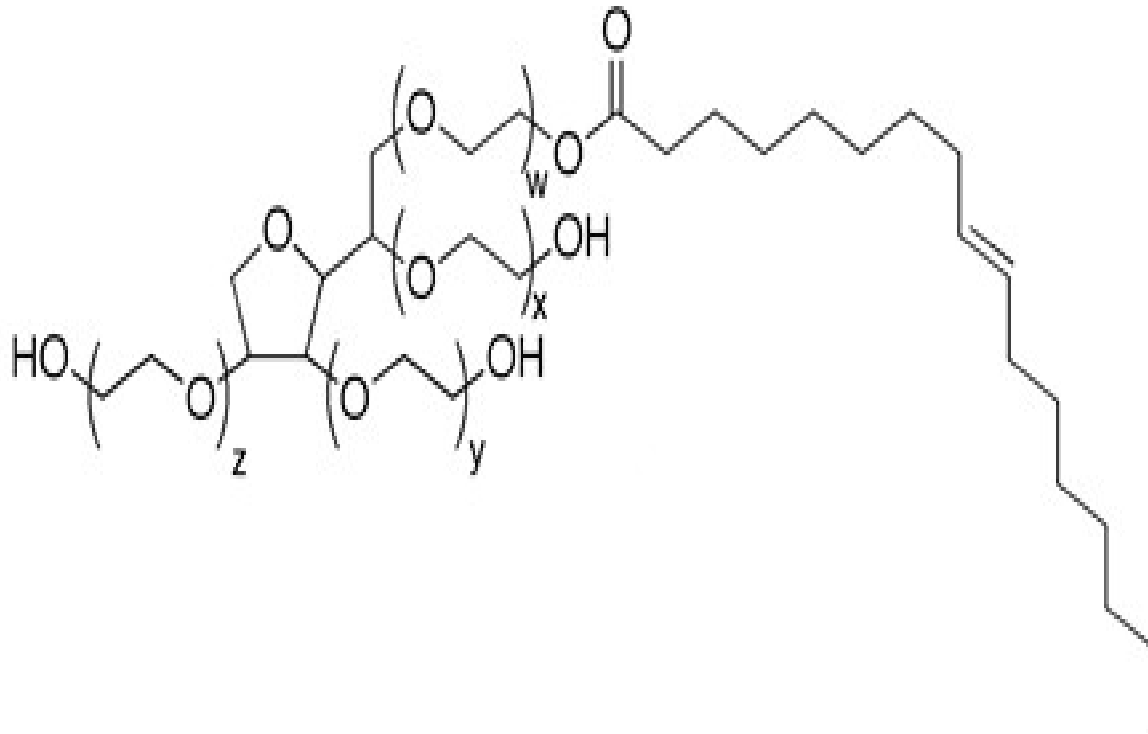
sorbitan



sorbitan monostearate

Polysorbates(Tweens®)

- Polysorbates are esters of polyoxyethylene sorbitan and fatty acids.



$$W+X+Y+Z= 4 \text{ or } 5 \text{ or } 20.$$

HLB calculations...

- ❑ For polysorbates (Tweens®) and sorbitan esters (Spans®)

$$\text{HLB} = \frac{E + P}{5}$$

- *E* is the percentage by weight of oxyethylene chains
- *P* is the percentage by weight of polyhydric alcohol group (sorbitan)
 - This formula can be also used to determine HLB of beeswax and lanolin derivative

HLB calculations...

- Alternatively, HLB value can be calculated (Davies formula)

$$HLB = 7 + \sum (\text{hydrophilic group numbers}) - \sum (\text{lipophilic group numbers})$$

Table 6.5 Group contributions to HLB values			
Group	Contribution	Group	Contribution
SO ₄ Na	+38.7	COOK	+21.1
COONa	+19.1	SO ₃ Na	+11.0
N (tertiary amine)	+9.4	Ester (sorbitan ring)	+6.8
Ester (free)	+2.4	COOH	+2.1
OH (free)	+1.9	-O-(ether)	+1.3
OH (sorbitan)	+0.5	CH, CH ₂ etc	
CF ₂ , CF ₃	-0.870	(alkyl)	-0.475
OCH ₂ CH ₂	+0.33	OCH(CH ₃)CH ₂	-0.15

Application of HLB

- HLB is used for selecting the appropriate quantities and types of emulsifiers during formulation of emulsions.
- In order to formulate a satisfactory emulsion, a blend of an oil-soluble emulsifying agent with a water-soluble emulsifying agent are used.

Application of HLB

- In addition to HLB of surfactants, which is required to formulate stable emulsion, another concept known as required HLB is needed.
- Required HLB value is the HLB value required by a particular hydrophobic component if it is to be emulsified effectively.

Required HLB value.....

- Each type of oil used will require an emulgent of a particular HLB number in order to ensure a stable product.
- The required HLB value differs depending on whether the final emulsion is O/W or W/O type.
 - For example, rHLB of liquid petrolatum is 4 for W/O emulsion and 10.5 for O/W emulsion.

TABLE 15-6**REQUIRED HYDROPHILIC-LIPOPHILIC BALANCE (HLB)
FOR SOME OIL-PHASE INGREDIENTS FOR OIL-IN-
WATER (O/W) AND WATER-IN-OIL (W/O) EMULSIONS***

	O/W	W/O
Cottonseed oil	6-7	—
Petrolatum	8	—
Beeswax	9-11	5
Paraffin wax	10	4
Mineral oil	10-12	5-6
Methyl silicone	11	—
Lanolin, anhydrous	12-14	8
Carnauba wax	12-14	—
Lauryl alcohol	14	—
Caster oil	14	—
Kerosene	12-14	—
Cetyl alcohol	13-16	—
Stearyl alcohol	15-16	—
Carbon tetrachloride	16	—
Lauric acid	16	—
Oleic acid	17	—
Stearic acid	17	—

*From *The Atlas HLB System*, Atlas Chemical Industries, Wilmington, DE; P. Becher, *Emulsions, Theory and Practice*, 2nd Ed., Reinhold, New York, 1966, p. 249.

Required HLB value.....

- Commonly an oil-soluble emulsifying agent with a water-soluble one produces the most satisfactory emulsions.
- The HLB of a mixture of surfactants, consisting weight ***fraction of A*** (x) and weight ***fraction of B*** ($1 - x$), is assumed to be an algebraic mean of the two HLB numbers:

$$\text{HLB}_{\text{mix}} = x \text{HLB}_A + (1-x) \text{HLB}_B$$

Example

- Calculate the weight of each surfactant required to prepare the following O/W emulsion:

Liquid petrolatum (rHLB 10.5).....50 g
Emulsifying agents.....5 g
Water.....100 g

the emulgents to be used are sorbitan monooleate (**HLB 4.3**) and polyoxyethylene sorbitan monooleate (**HLB 15**)

Adsorption at solid interfaces

- Adsorption of material at solid interfaces can take place either from adjacent liquid or gas phases.
- The study of adsorption of gases on solids arises in such diverse applications as **the removal of objectionable odors from rooms and foods, the operation of gas masks**, and the **measurement of surface area of particles in a powder**.

Adsorption at solid-gas interfaces

- The degree of adsorption of a gas by a solid depends on:
 - ✓ the **chemical nature of the adsorbent**(the material used to adsorb the gas)
 - ✓ the **chemical nature of the adsorbate.**
 - ✓ the **surface area of the adsorbent,**
 - ✓ **the temperature**
 - ✓ **partial pressure of the adsorbed gas.**

Adsorption at solid-gas interfaces (cont'd)

- ✓ Physically adsorbed gas can be desorbed from a solid by increasing temperature and reducing the pressure.

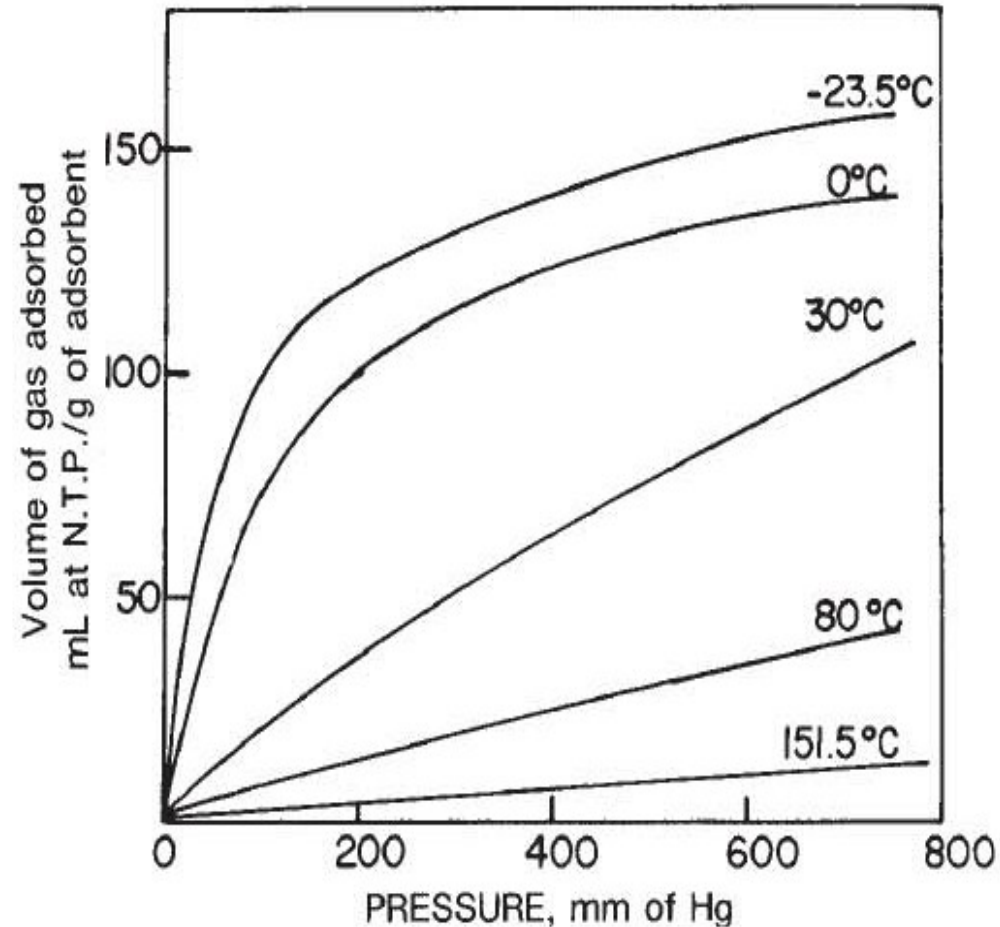


Figure 20-9. Adsorption isotherms for ammonia on charcoal. (From Titoff Z. Z Phys Chem Leipzig 1910; 74: 641.)

Adsorption at solid-gas interfaces

- There are two types of adsorption at solid gas interfaces:
 - ✓ **Physical adsorption or van der Waals adsorption**
 - ✓ **Chemical adsorption (chemisorption)**
- Physical adsorption of gases on solid occur at temperatures approaching the liquefaction temperature of gases.

Adsorption at solid-gas interfaces (cont'd)

- Physical adsorption is **reversible**.
- The removal of the adsorbate from the adsorbent is known as **desorption**.
- **Chemisorption** in which the adsorbate is attached to the adsorbent by primary chemical bonds, is irreversible unless the bonds are broken.

Adsorption at solid-gas interfaces (cont'd)

- The relationship between the amount of gas adsorbed by a solid at a constant temperature (T) as a function of the gas pressure (P) is defined as *adsorption isotherm*.
- The term **isotherm** refers to a plot at constant temperature.

Adsorption at solid-gas interfaces (cont'd)

- The number of **moles** or **grams** or **milliliters**, **x**, of gas adsorbed on, **m**, **grams of adsorbent** at constant temperature is plotted on the vertical axis against the equilibrium pressure of the gas in mmHg on the horizontal axis is shown on the next slide.

Adsorption at solid-gas interfaces (cont'd)

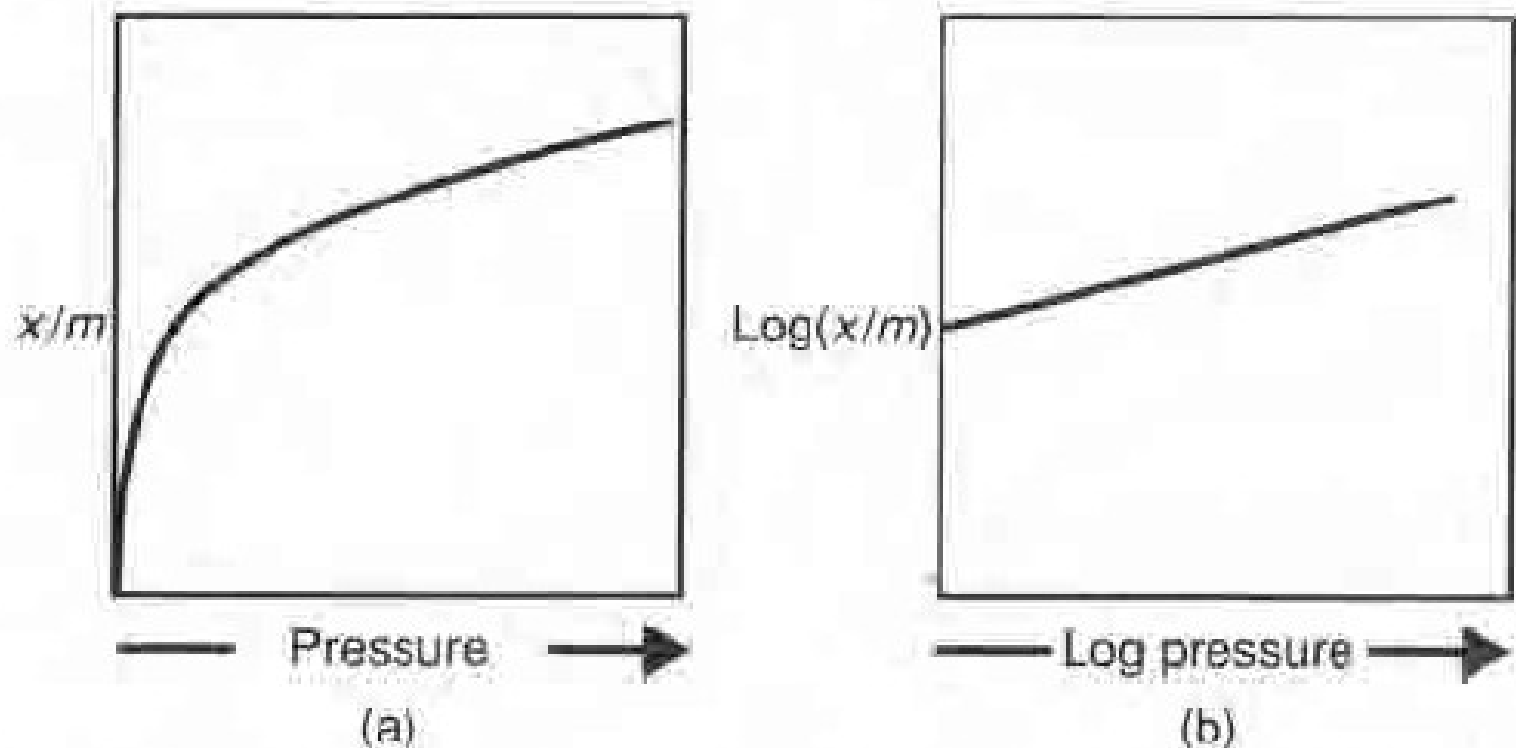
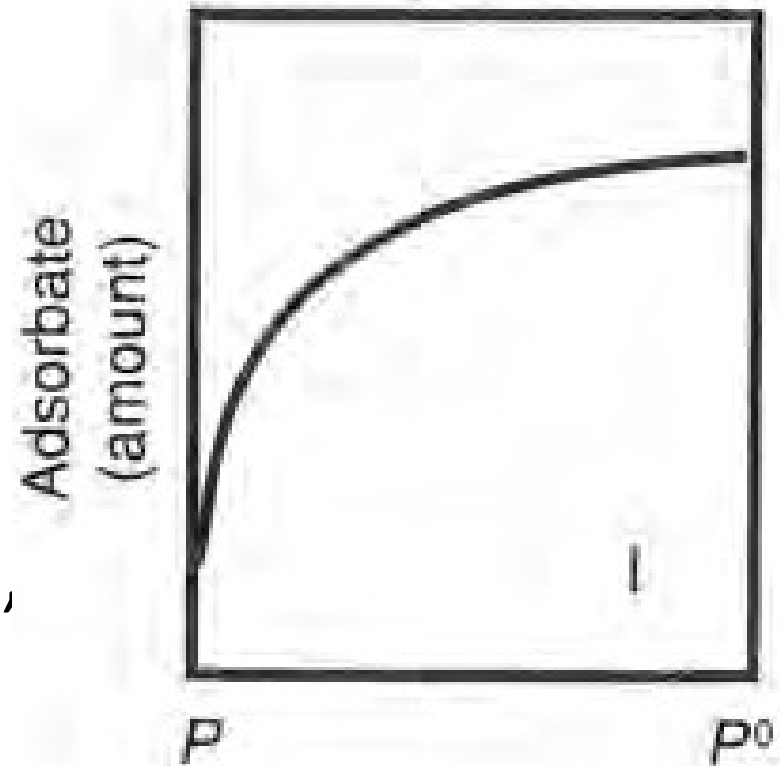


Fig. Adsorption isotherms for a gas on a solid. (a) Amount, x , of gas adsorbed per unit mass, m , of adsorbent plotted against the equilibrium pressure. (b) Log of the amount of gas adsorbed per unit mass of adsorbent plotted against the log of the pressure.

Adsorption at solid-gas interfaces (cont'd)

- Adsorption significantly increases with increasing pressure, followed by leveling-off which is due to either to a saturation of available specific chemical groups, as in chemisorption, or to the entire available surface being covered by physically adsorbed molecules.



Adsorption at solid-gas interfaces

(cont'd)

- In the case of physical adsorption after adsorption levels-off, often a marked increase in adsorption occurs, presumably due to multilayered adsorption.

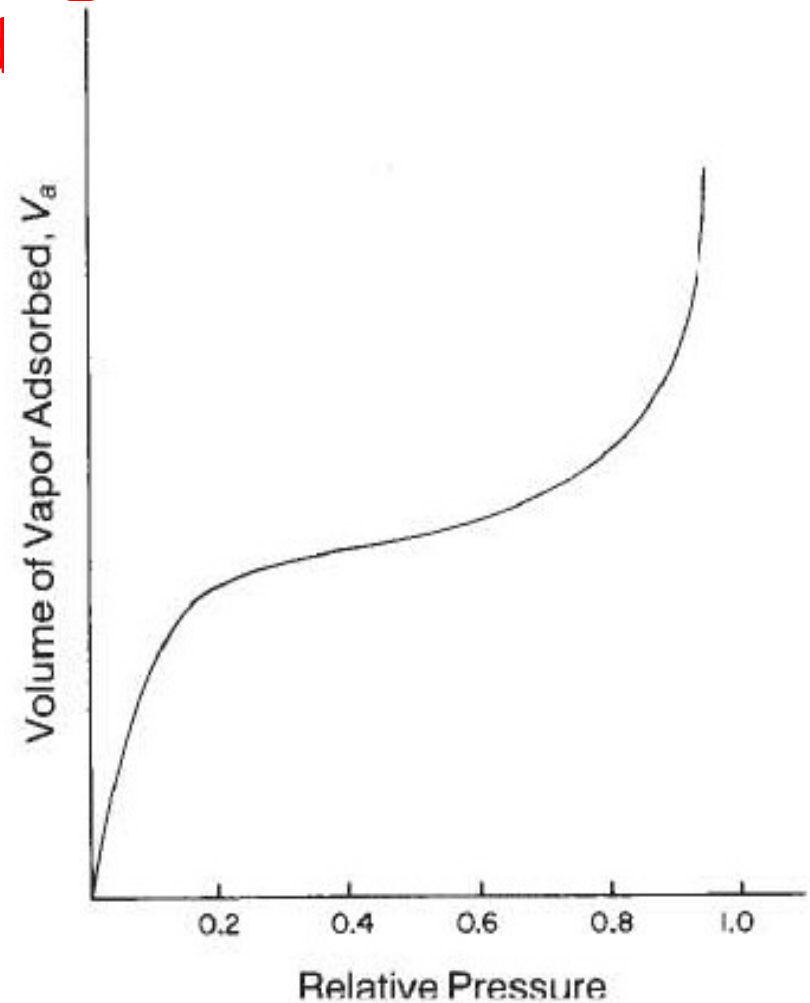


Figure 20-10. Typical plot for multilayer physical adsorption of a vapor on a solid surface.

Adsorption at solid-gas interfaces (cont'd)

- **Freundlich** suggested a relationship, the *Freundlich isotherm*,

$$y = \frac{x}{m} = KP^n$$

where

- ✓ **y** is the mass of gas, **x**, adsorbed per unit mass, **m**, of adsorbent
- ✓ **K** is constant reflecting adsorption affinity of the gas to the solid.
- ✓ **n** is constant reflecting adsorption capacity of the solid

Adsorption at solid-gas interfaces (cont'd)

- The equation is handled more conveniently when written in logarithmic form:

$$\log \frac{x}{m} = \log k + n \log p$$

Adsorption at solid-gas interfaces (cont'd)

- **Langmuir** postulated that for adsorption to occur, a solid must contain uniform adsorption sites, each capable of holding a single gas molecule.
- Molecules colliding with the surface may bounce off elastically or they may remain in contact for a period of time.
- Molecular contact on the solid surface over the period of time was termed adsorption by Langmuir.

Adsorption at solid-gas interfaces (cont'd)

- Two major assumptions were made in deriving Langmuir adsorption equation.
 - ✓ Only those molecules striking an empty site can be adsorbed; hence, only monomolecular adsorption occurs.
 - ✓ The forces of interaction between adsorbed molecules are negligible, and therefore, the probability of a molecule adsorbing onto or desorbing from any site is independent of surrounding sites.

Adsorption at solid-gas interfaces (cont'd)

- The fraction of centers occupied by gas molecules at pressure P is represented by θ , and fraction of sites not occupied is $1 - \theta$.
- The rate of adsorption (r_1) or condensation of gas molecules on the surface is proportional to the unoccupied spots, $1 - \theta$, and to the pressure, P ,

$$r_1 = K_1(1 - \theta)p \dots \dots \dots 1$$

Adsorption at solid-gas interfaces (cont'd)

- The rate of evaporation(desorption) (r_2) of gas molecules bound on the surface is proportional to the fraction of surface occupied, θ , or

$$r_2 = k_2\theta \text{2}$$

- At equilibrium, $r_1 = r_2$ or

$$K_1(1-\theta)p = k_2\theta \text{3}$$

Adsorption at solid-gas interfaces (cont'd)

- By rearrangement, we obtain

$$\theta = \frac{K_1 P}{K_2 + P K_1} = \frac{[K_1/K_2] P}{1 + P [K_1/K_2]} \dots\dots\dots 4$$

Adsorption at solid-gas interfaces (cont'd)

- We can replace K_1/k_2 , by b and θ by v/v_m .
- Inserting these terms into equation (4) produces the formula:

$$v = \frac{v_m b p}{1 + b p} \dots\dots\dots 5$$

- ✓ V is volume of gas adsorbed per gram of adsorbent at pressure P
- ✓ V_m is the volume of gas adsorbed per gram of adsorbent when the whole surface of adsorbent is covered by monomolecular layer
- ✓ b is a constant indicating affinity of adsorbate for the adsorbent .

Adsorption at solid-gas interfaces (cont'd)

- By inverting equation (5), and multiplying through by p , we can rewrite as follows:

$$\frac{P}{V} = \frac{1}{bV_m} + \frac{P}{V_m} \dots\dots\dots 6 \text{ (Langmuir isotherm)}$$

- A plot of p/v against p yield a straight line, and v_m and b can be obtained from the slope and intercept.

Multilayered adsorption

- A major assumption of the Langmuir isotherm model is that adsorption stops at monolayer coverage.
- For multilayer adsorption, Langmuir equation was modified by **Brunauer, Emmett and Teller** (**BET equation**)

$$\frac{p}{V(p_0 - p)} = \frac{1}{V_m b} + \frac{(b - 1)p}{V_m b p_0}$$

Multilayered adsorption

$$\frac{p}{V(p_0 - p)} = \frac{1}{V_m b} + \frac{(b - 1)p}{V_m b p_0}$$

Where

- ✓ **P** is pressure of the adsorbate at which the volume, **V** of vapor per gram of adsorbent is adsorbed.
- ✓ **V_m** is the quantity of the vapor adsorbed per gram of adsorbent when the surface of adsorbent is covered with monomolecular layer.

Multilayered adsorption

- ✓ b is a constant proportional to the difference between the heat of adsorption of the gas in the first layer and the latent heat of condensation of successive layers.
- ✓ p_0 is vapor pressure when the adsorbent is saturated with adsorbate.
- The saturation vapor pressure p_0 is obtained by bringing excess adsorbate in contact with the adsorbent.

Surface area of solid calculation by BET isotherm

- One of the most common uses of the BET isotherm is for determining the surface area of finely divided solids by physical adsorption.
- The monolayer capacity V_m is of primary interest because it allows for the calculation of the surface area based on the area occupied by each adsorbed gas molecule.

Surface area of solid calculation by BET isotherm

- The adsorbate most commonly employed for BET surface area determinations is **nitrogen**, which has an effective area per molecule (A) of 0.162 nm^2 at liquid nitrogen temperature (77 K).
- Other useful gases include argon ($A = 0.138 \text{ nm}^2$) and krypton ($A = 0.195 \text{ nm}^2$).

Surface area of solid determination by BET method

- Plot of $\frac{p}{v(p_0 - p)}$ against $\frac{p}{p_0}$ gives a straight line.
- The slope and intercepts enables to obtain the values of b and v_m
- The equation is linear only in the range $0.05 \leq \frac{p}{p_0} \leq 0.35$.

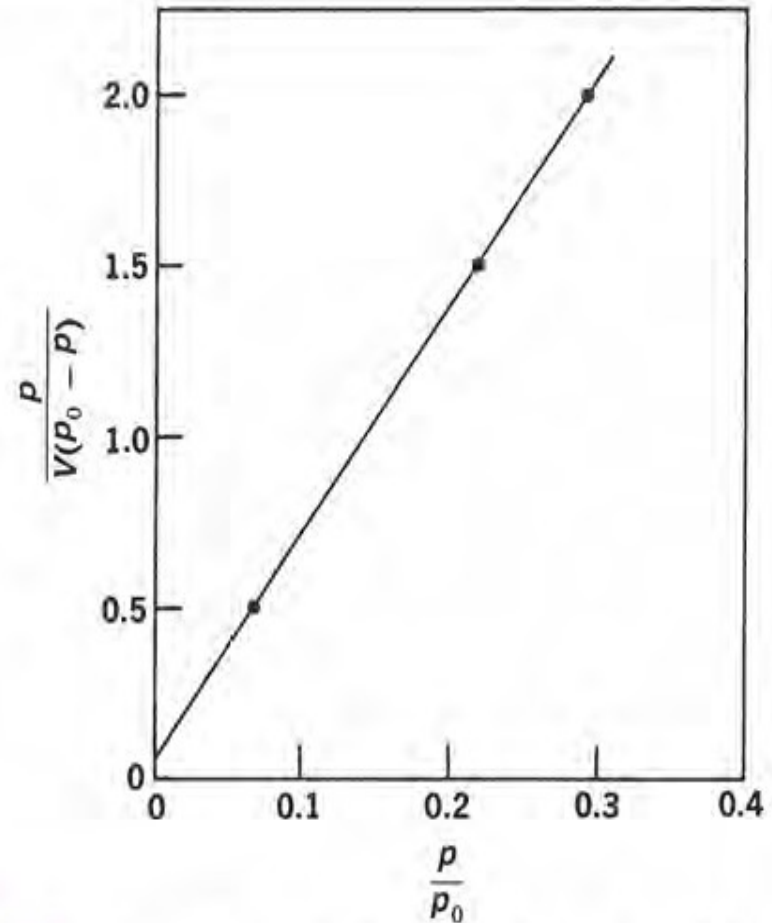


Fig. 18-11. A linear plot of the Brunauer, Emmett, and Teller (BET) equation for the adsorption of nitrogen on a powder.

Surface area of powder determination by BET method

$$\frac{p}{V(p_0 - p)} = \frac{1}{V_m b} + \frac{(b-1)p}{V_m b p_0}$$

- Slope (S) = $(b-1)/bV_m$

- Intercept (I) = $1/bV_m$

- After substitution to eliminate **b**,

$$V_m = \frac{1}{S + I}$$

Surface area of solid determination by BET method

- Surface area of the adsorbent (S) will be

$$S = \frac{A \times N_A}{M_v} \times v_m$$

Where,

- ✓ S is surface area of adsorbent (solid)
- ✓ A is effective area of single gas molecule
- ✓ N_A is Avogadro's number (6.02×10^{23})
- ✓ M_v is molar volume of gas at STP (22.4 liters)
- ✓ V_m is volume of gas adsorbed per gram of adsorbent to form monomolecular layer

Surface area of solid determination by BET method

- For nitrogen as adsorbate,

$$S = \frac{0.162 \text{ nm}^2 \times 6.02 \times 10^{23}}{22.4 \text{ L}} \times v_m$$

$$S = 4.35 \text{ m}^2 / \text{cm}^3 \times v_m$$

Surface area of solid determination by BET method

- Example, the adsorption of nitrogen on silica gel at 90 K in the p/p_o region of 0.05 to 0.35 is shown in the table below. Calculate the surface area of silica gel according to the BET isotherm.

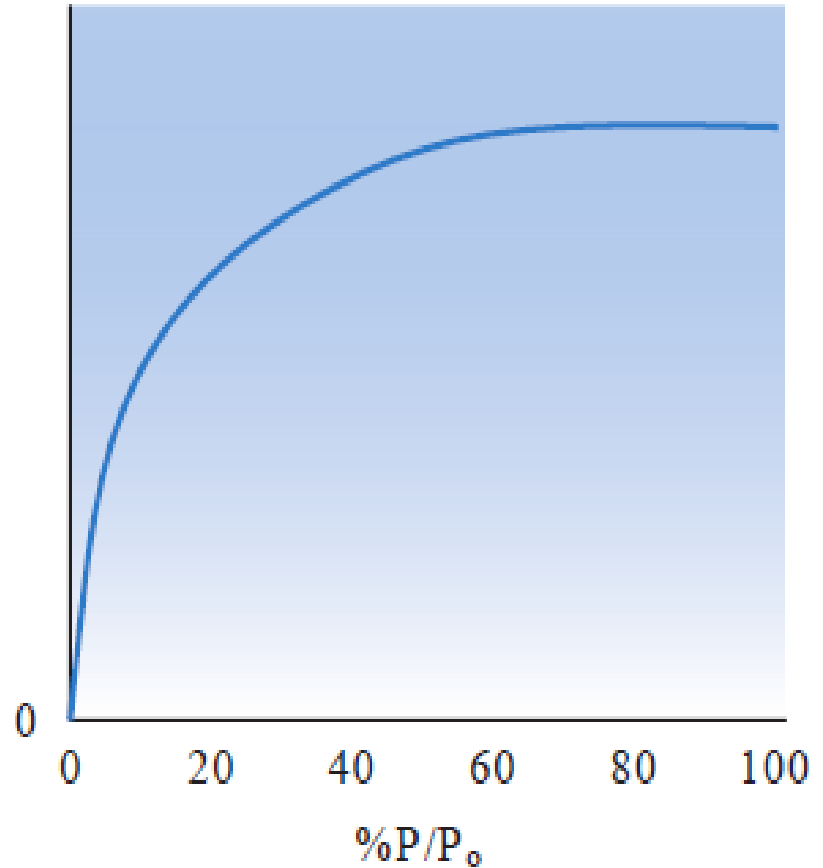
$V(\text{cm}^3/\text{g}),$	39	41	44	48	50	53	58	61
p/p_o	0.05	0.07	0.10	0.16	0.20	0.25	0.30	0.35

Adsorption isotherms

- There are five common types of adsorption isotherms:
 - ✓ **Type I isotherms (Langmuir isotherm)**
 - ✓ **Type II isotherms**
 - ✓ **Type III isotherms**
 - ✓ **Type IV isotherms**
 - ✓ **Type V isotherms**

Type I isotherms

- ✓ Usually termed as Langmuir isotherm.
- ✓ Applicable for monomolecular adsorption.
- ✓ Rapid initial rise in the amount of gas adsorbed with increasing gas pressure until some limiting value is reached.



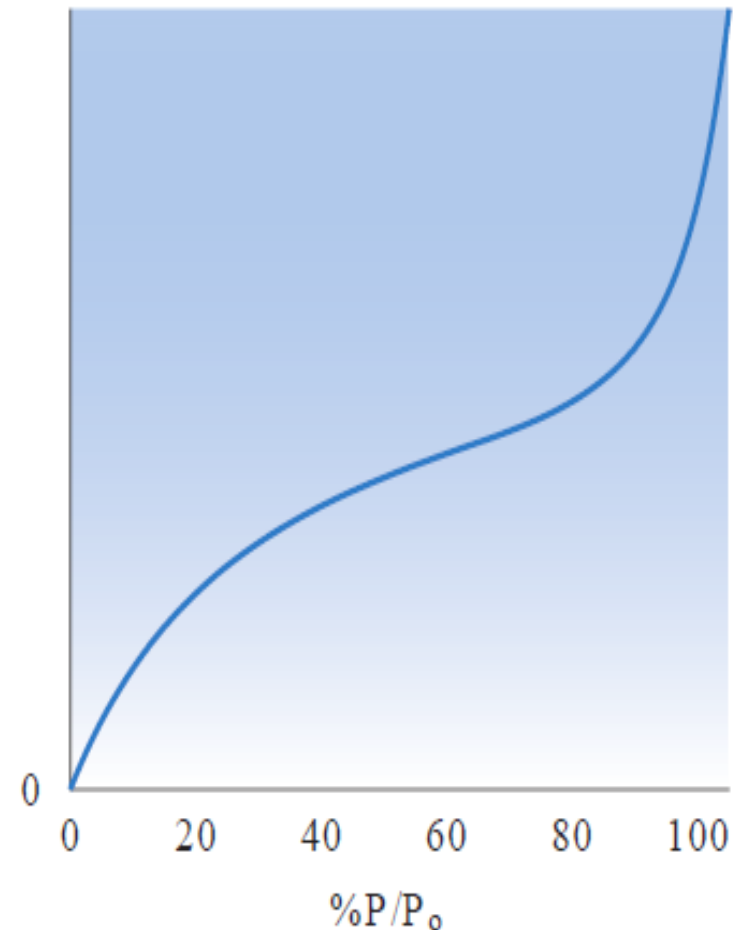
Type I isotherms

- ✓ The limiting value is attained when the entire surface is covered by monomolecular layer.
- ✓ **Strong non-specific attractive interaction between the adsorbate and adsorbent.**
- ✓ **Weak attraction between the adsorbate molecules themselves.**

Type II isotherms

- ✓ Typical of physical adsorption on nonporous solids.
- ✓ Form multimolecular layer.
- ✓ Adsorbate molecules have relatively strong mutual interactions .
- ✓ Most common type of isotherm.
- ✓ Described by BET equation.

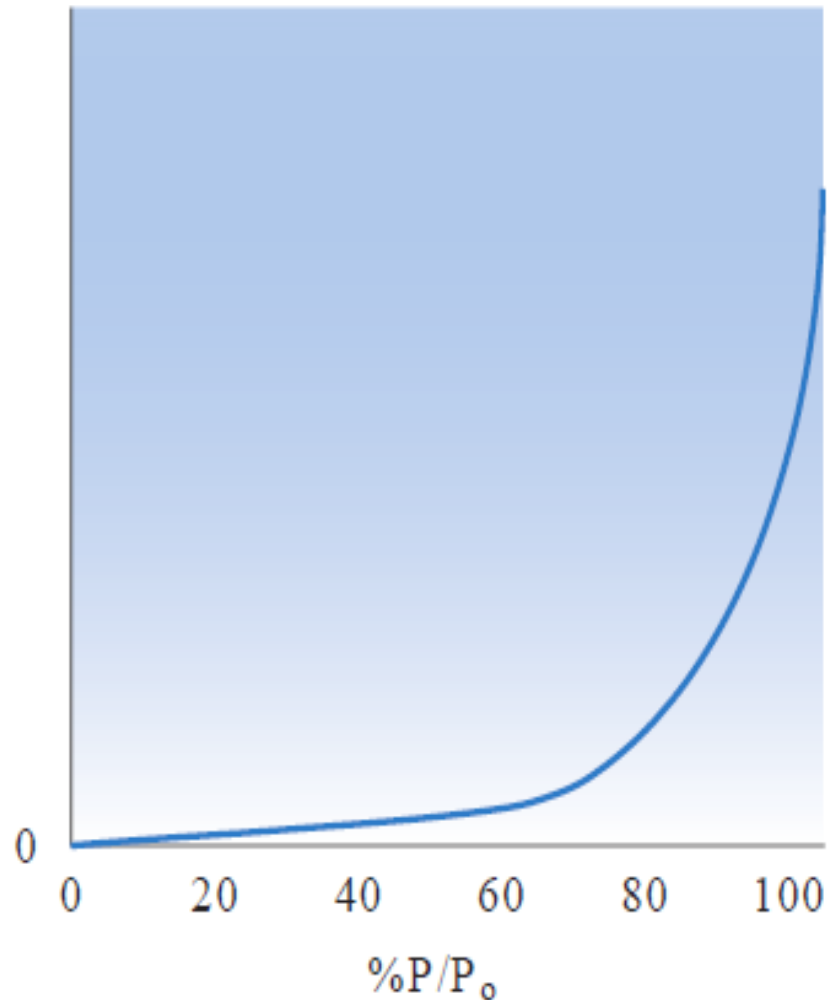
Type II



Type III isotherms

Type III

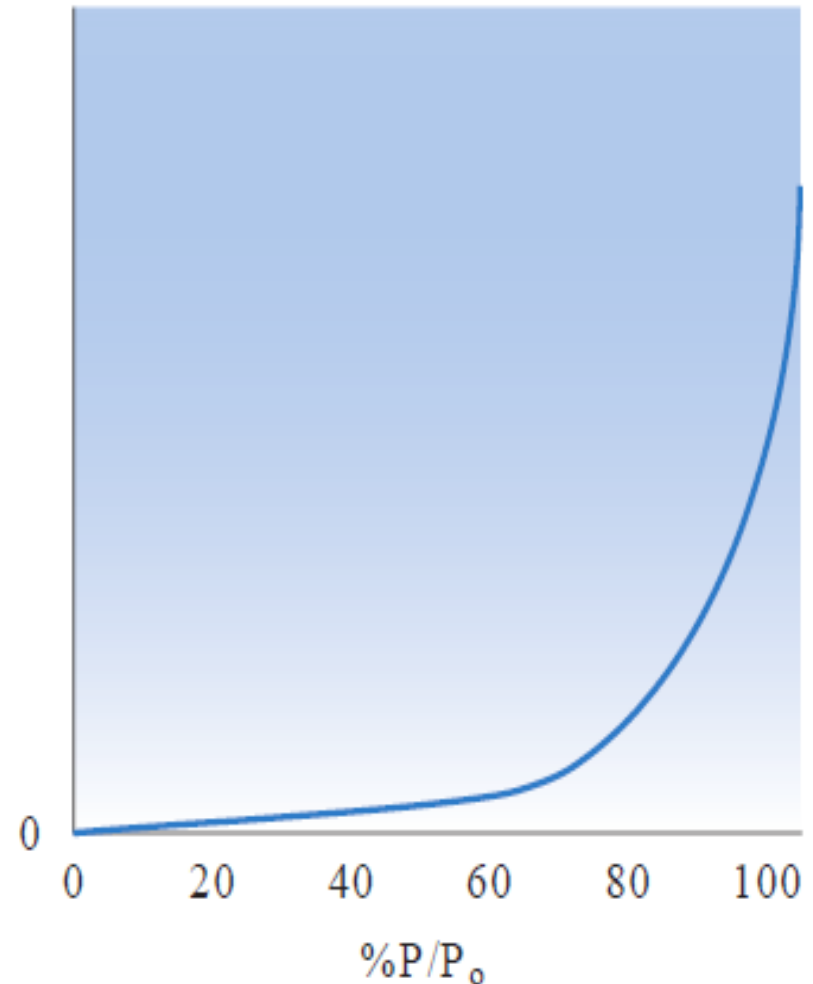
- ✓ The adsorbate and adsorbent have no great affinity for each other.
- ✓ Greater attractive interaction between adsorbate molecules.



Type III isotherms

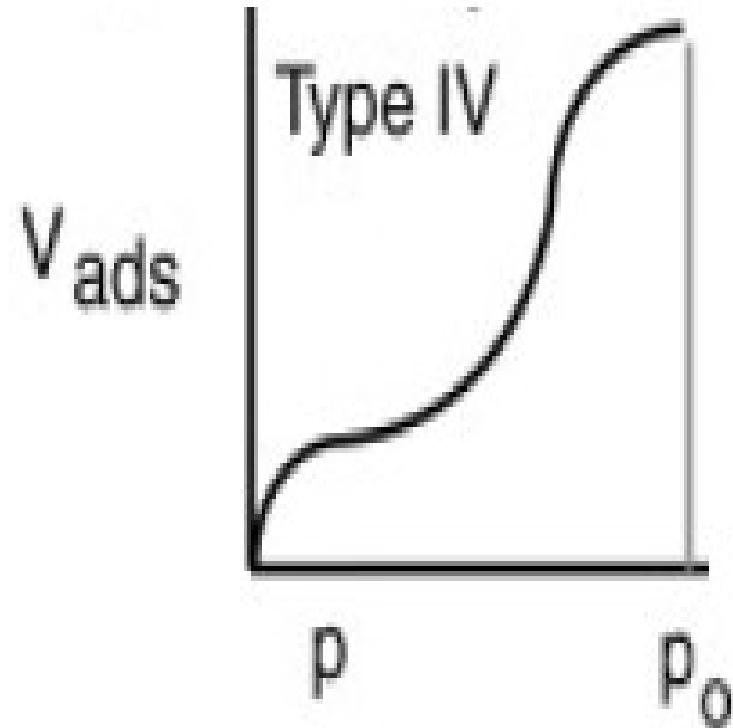
- ✓ Significant pressure is required to effect adsorption initially.
- ✓ Later, favorable adsorbate –adsorbate interaction favors for rapid adsorption of the gas.
- ✓ Rare type of isotherm.

Type III



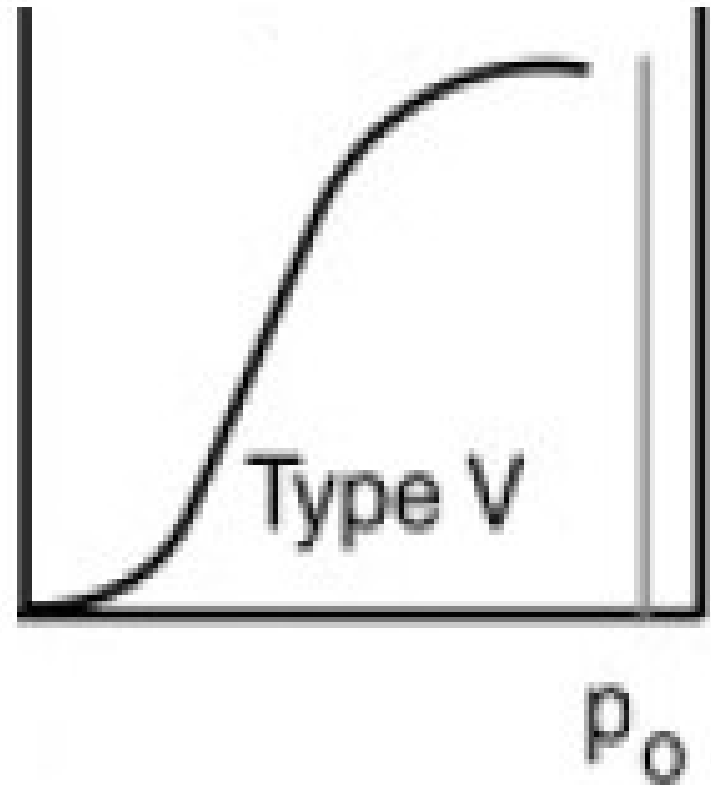
Type IV isotherms

- ✓ Similar to Type II isotherms and corresponds to systems involving capillary condensation in porous solids.



Type V isotherms

- ✓ Similar to Type III, are rare type of isotherm.
- ✓ The adsorbate and adsorbent have no great affinity for each other.
- ✓ Solids have porous structure.



Adsorption at solid-liquid interface

- Drugs can be adsorbed from solutions on to solids such as activated charcoal and alumina.
- The adsorption of solute molecules from solution generally leads to a monomolecular layer and often described by Langmuir equation.

Adsorption at solid-liquid interface.....

$$\frac{C}{y} = \frac{1}{by_m} + \frac{C}{y_m}$$

Where,

- ✓ C is the equilibrium concentration of adsorbate in mgs per 100 mL of solution.
- ✓ y is the amount of adsorbate, x, in mgs adsorbed per gram, m, of solid (i.e., $y=x/m$).
- ✓ b is affinity of the adsorbate for the solid.
- ✓ y_m is the amount of adsorbate in mg adsorbed per gram of adsorbent when the monolayer is complete.

Adsorption at solid-liquid interface

- A solvent may compete with a solute for adsorption on solid surfaces.
- Solvents also affect escaping tendency of solutes. For example, the more polar solute molecule is, the less adsorption occurs from water.
 - ✓ The adsorption of various fatty acids from water on charcoal increases with increasing alkyl chain length or non-polarity.

Adsorption at solid-liquid interface

- ✓ The adsorption of various fatty acids from water on charcoal increases with increasing alkyl chain length or non-polarity.

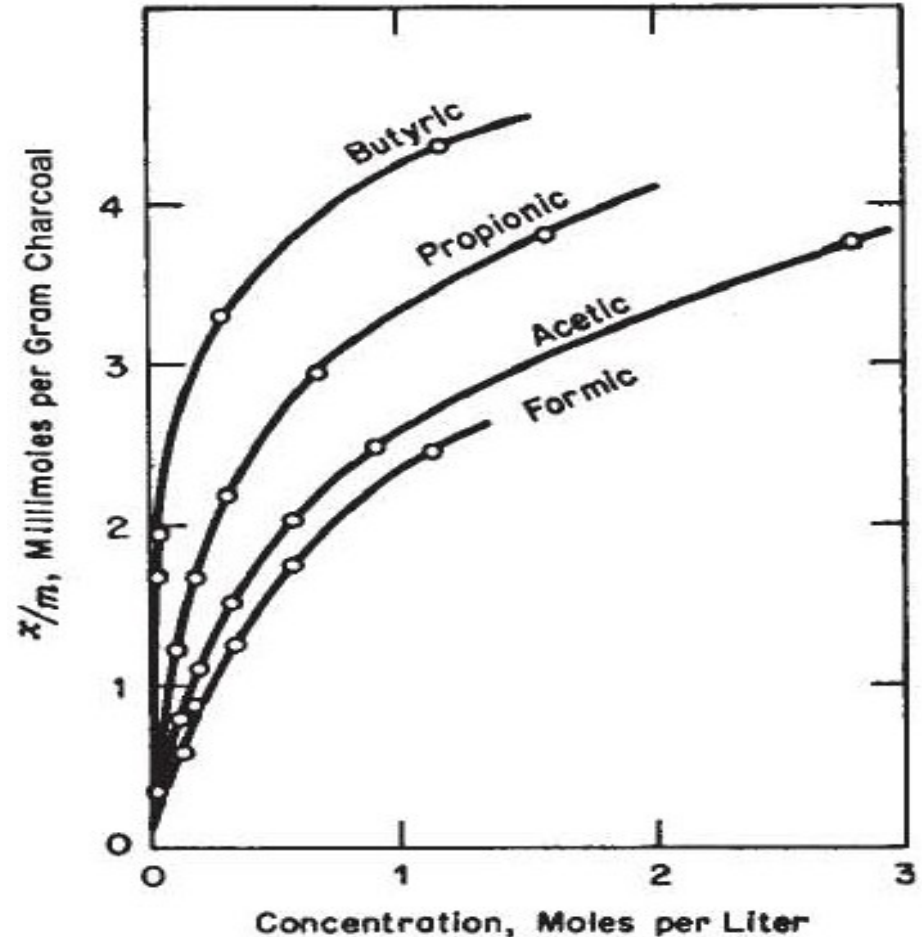


Figure 20-15. The relation between adsorption and molecular weight of fatty acids. (From Weiser HB. *A Textbook of Colloid Chemistry*. New York: Elsevier, 1949.)

Adsorption at solid-liquid interface

- In general, the more chemically unlike the solute and solvent and the more alike the solid surface groups and solute, the greater extent of adsorption.

Adsorption at solid-liquid interface

- ✓ Charged solid surfaces such as polyelectrolytes, will strongly adsorb oppositely charged solutes.
- ✓ Example, a surfactant, LN^+ adsorb with negatively charged silica or glass surface.

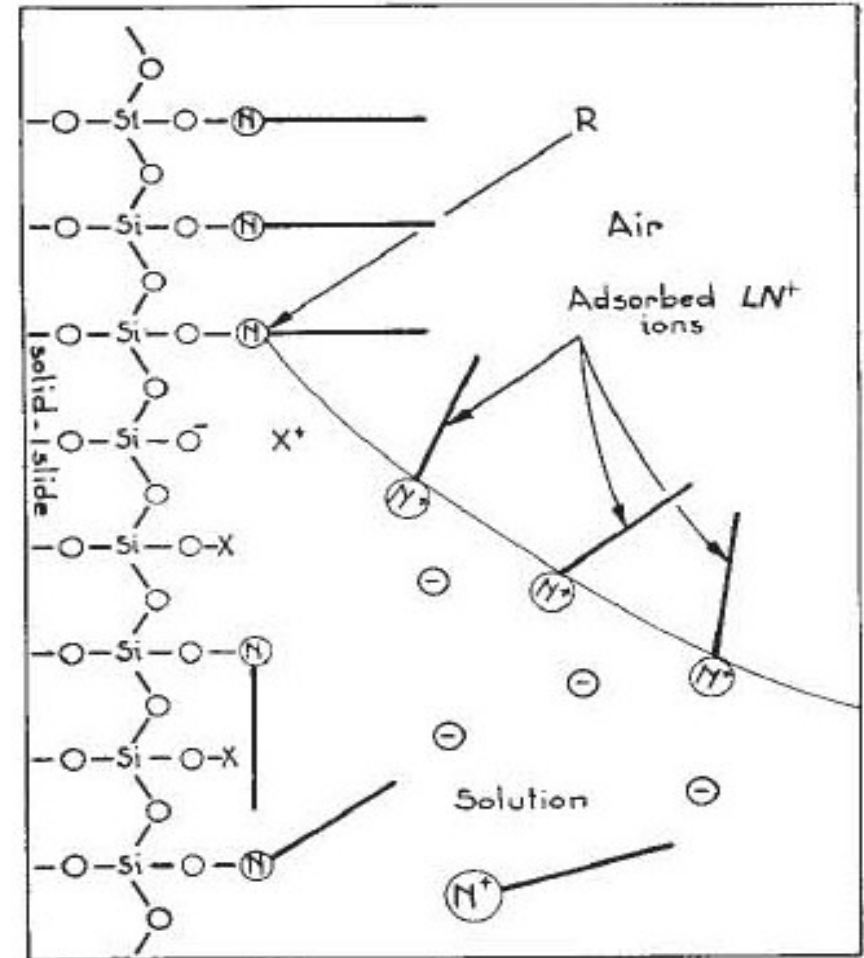


Figure 20-16. The adsorption of a cationic surfactant, LN^+ , onto a negatively charged silica or glass surface, exposing a hydrophobic surface as the solid is exposed to air. (From Ter-Minassian-Saraga L. *Adv Chem Ser* 1964; 43: 232.)